Practical 4: PGE petrography and volumetrics calculation for a PGE deposit

Part B

This calculation illustrates the basic simplicity of maths used in reserve/resource calculations. Note that a number of uncertainties such as ore losses are not considered here. Full evaluations would also take a statistical approach. Data modified from Grobler (2019): https://doi.org/10.1007/s00126-018-0800-x

Data:

Consider a prospect on the northern limb of the Bushveld Complex, exploring for PGEs and Cr. The extract from the geological map, block diagram and cross section on the next 2 pages (Figs. 1-3) show the stratigraphy, dip of strata, main faults, and exploration data (boreholes, cross sections). The prospect area is outlined in the red dashed line. Exploration focuses on several units, including the Tennis Ball Marker (TBM, shown in blue on the map and block diagram).

Although in reality the rocks are folded, for this exercise consider the TBM as a simple sheet of rock with a thickness of 0.8 m.

The reef consists of a pegmatitic pyroxenite with the following modal composition:
70 wt% pyroxene (diopside, $CaMgSi_2O_6$)
25 wt% plagioclase (anorthite, $CaAl_2Si_2O_8$)
2 wt% sulfide (pyrrhotite, $Fe_{1-x}S$)
3 wt% chromite, $FeCr_2O_4$

Questions:

1) What is the total resource of platinum in this prospect?

2) The resource calculated is for total PGEs – assuming 1/3 of this by weight is Pt, and considering the cost of extracting an ounce of Pt is 1000$, check current market prices for Pt to see if mining is economically viable. What parameters can change around economic viability?

3) Calculate the expected temperature in a mine shaft at 1km TVD. A map of geothermal gradients is attached (Figs. 4, 5).

Homework: What is the total resource down to 1 km vertical depth (typical Bushveld mine depth)?

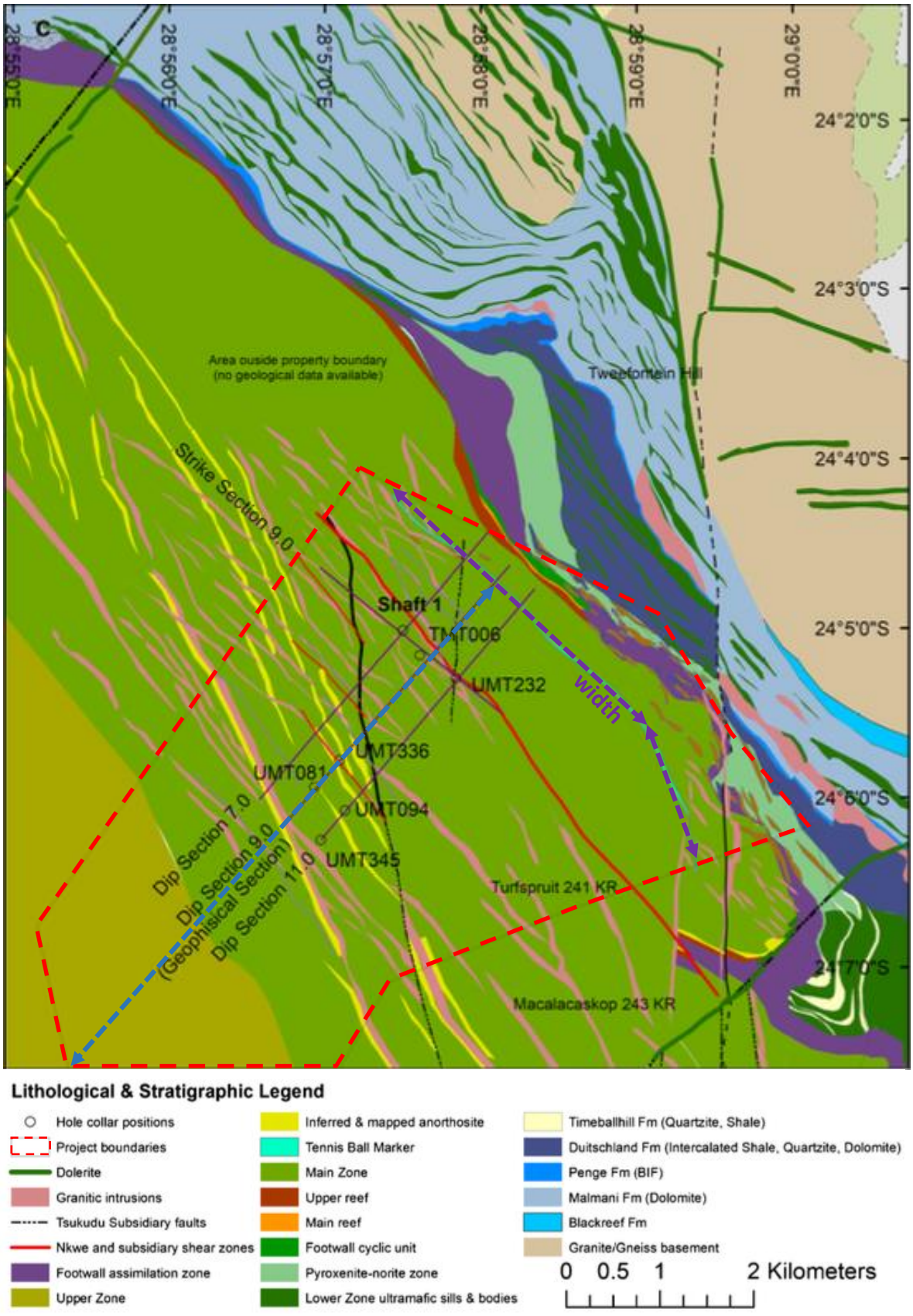


Fig. 1: Geological map (Grobler 2019)

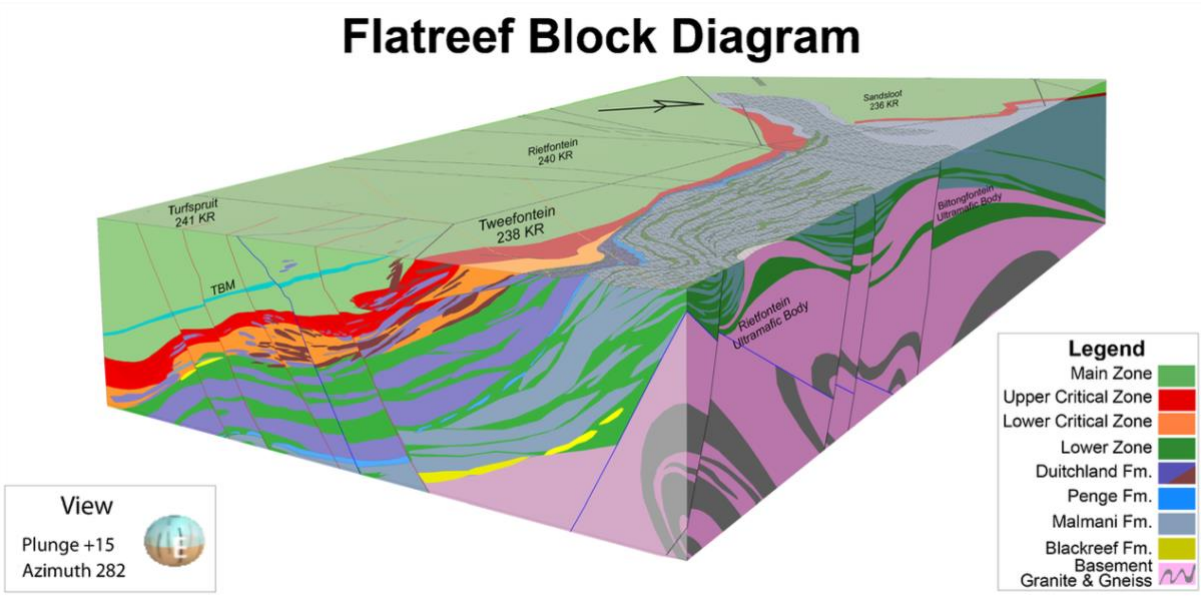


Fig. 2: Block diagram (Grobler 2019)

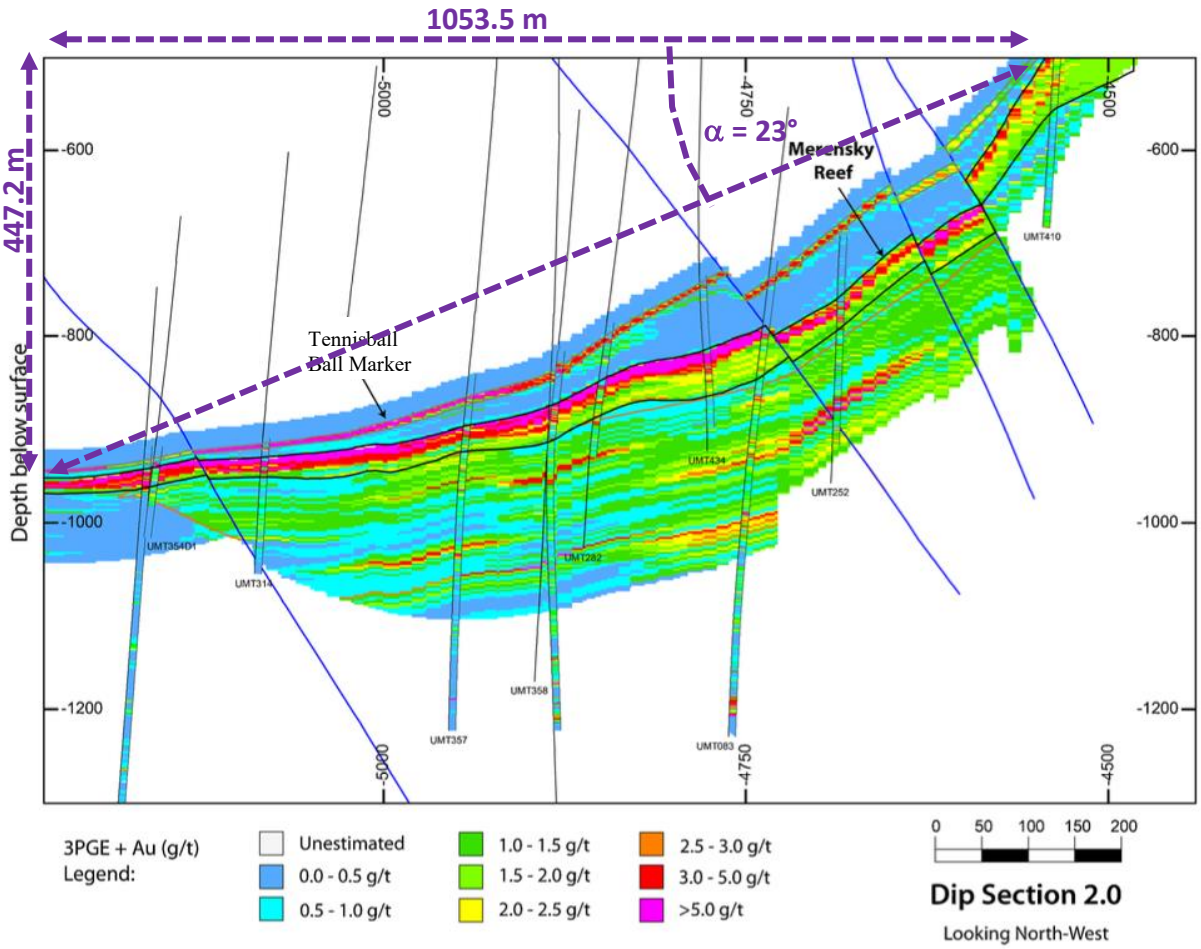


Fig. 3: Cross section of part of the prospect with borehole-based grades (Grobler 2019)

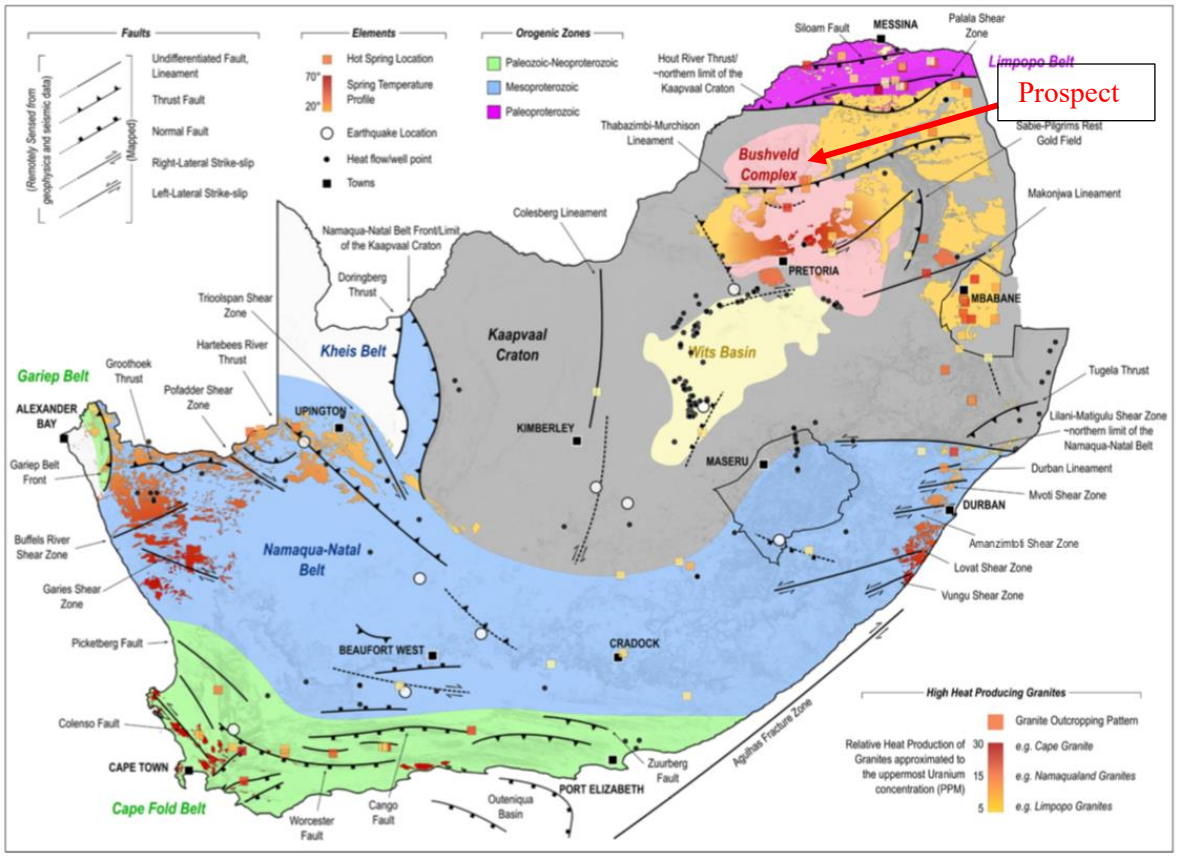


Fig. 4: Location map of the Bushveld Complex and the mine (Dhansay 2017).

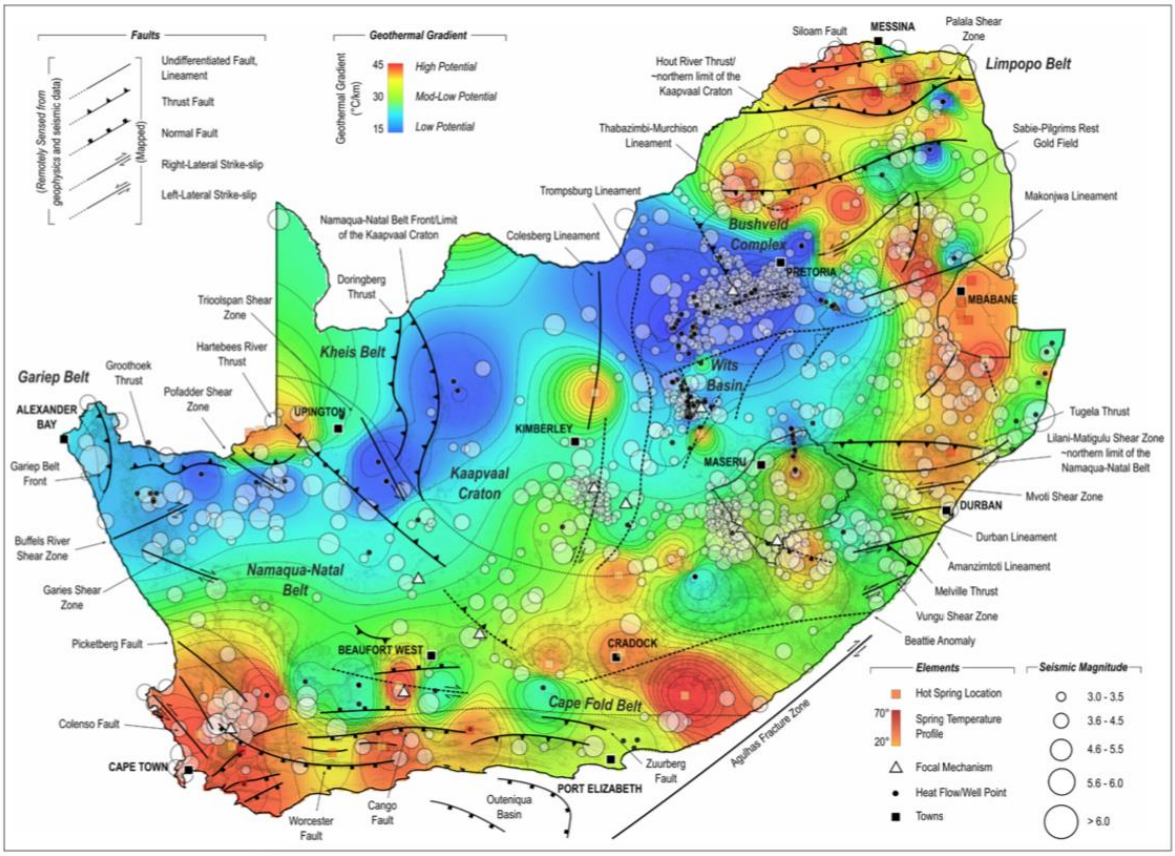


Fig. 5: Map pf geothermal gradient (Dhansay 2017).

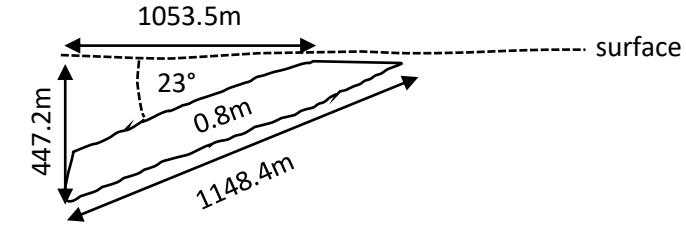
Solution:
1. calculate rock volume:
a. thickness is known: 0.8 m

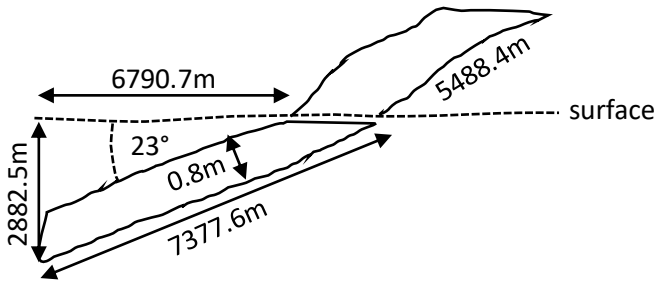
b. width: measure along outcrop section (can break it down into segments for more precision, or measure straight between prospect boundaries
width (measured straight) = 5488.4 m

c. length along dip
first determine the dip angle from cross section, or calculate using trigonometry
$\alpha = 23°$
$\tan \alpha = 447.2 / 1053.5$

d. length along depth to prospect boundary in SW (following dip section 9.0):
horizontal distance from outcrop: 6790.7 m
dip length at this point: $\cos 23 = 6790.7 / x \Leftrightarrow x = 6790.7 / \sin 23 = 7377.6$ m
dip length = 7377.6 m

e. **volume:** 7377.6 x 5488.4 x 0.8 = 32392846.2 m^3 of rock = **32.4×10^6 m^3**





2. convert to mass
Need to determine bulk density of reef (find information via mindat.org):
Pyroxene: density = 3.3 $g/cm^3 = t/m^3$
Plagioclase: density = 2.75 $g/cm^3 = t/m^3$
Sulfide: density = 4.62 $g/cm^3 = t/m^3$
Chromite: density = 4.65 $g/cm^3 = t/m^3$
Reef density: 70x3.3 + 25x2.75 + 2x4.62 + 3x4.65 = 3.23 $g/cm^3 = t/m^3$
Mass of reef: 32392846.2 m^3 x 3.23 t/m^3 = 104628893.2 t = 104.6 10^6 t rock

3. resources
Take a grade from fig 3: range 3-5 g/t, eg 4 g/t
104628893.2 t x 4 g/t = 418515572.8 g PGE= 418.5 t PGE
1/3 is Pt: 13950519.1 g Pt = 4.9 million oz (1g = 0.035274 oz)